

CORE APPROACH

AI for perception + visualization.

Deterministic code for products + budget.

The problem Most bathroom AI stops at a pretty image: it can hallucinate products, duplicate fixtures and ignore the budget.

1

AI MODEL

Understand

Gemini reads only visible evidence and returns structured JSON - no guessed dimensions or brands.

2

DETERMINISTIC CODE

Select

A rule-based recommender chooses real catalogue products within budget. AI never sets a product or a price.

3

AI MODEL

Generate

Stability redesigns the uploaded room. Pollinations renders new spaces. Prompt guardrails control fixture counts.

OUTCOME One system turns room context, budget and real products into explainable, photorealistic design decisions.

System architecture

One FastAPI backend separates exact business logic from probabilistic AI services.

CLIENT



React + Vite

- Photo upload
- Room, budget and style
- Product comparison
- Design results

SERVER

FastAPI backend

Deterministic engine

Recommender and budget explorer
Design assistant
25-SKU Kohler India catalogue

AI orchestration

Prompt builders + JSON parsing
Retries and model fallback
Upload validation
API keys stay server-side

AI SERVICES

Gemini
Analyzes your photo

Stability AI
Redesigns the same room

Pollinations Flux
Renders a new concept

Tech stack

A lightweight web stack with specialist AI models assigned one job each.



Frontend

Responsive design flow with client-side image preprocessing before upload.

React 19

Vite

Axios



Backend

FastAPI endpoints orchestrate deterministic business logic and AI services.

FastAPI

SQLAlchemy

Pillow



AI services

Vision, photo redesign and new-space rendering are isolated behind dedicated services.

Gemini

Pollinations

Stability AI



Reliability and safety

Retries, fallback models and upload validation. API keys remain server-side.

Backoff

Fallback

Validation

Innovation pitch

Why AI Atelier is a design decision engine - not another image generator.

1

Grounded redesign, not blind generation

Gemini reads the actual room first; those findings condition the redesign instead of starting from a generic prompt.

2

Catalogue-grounded, budget-aware

Real SKUs and prices come from the catalogue; Smart, Balanced and Elevated tiers make trade-offs explicit.

3

Creativity with hard guardrails

Fixture-count rules, validation and negative prompts constrain common AI errors such as duplicate sanitary fixtures.

4

A decision engine, not a moodboard

Both journeys connect visualization to product and budget logic, turning inspiration into an actionable design path.

fFUTURE ADDITIONS : Image-to-3D room reconstruction + automatic fixture-to-SKU matching.